Materials, stress & strain

National 5 · Mechanisms & structures · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ choose and justify a material from a data table
- ☐ calculate stress, strain and Young's modulus
- ☐ label a stress–strain graph: limit of proportionality, yield, UTS, break

I DO — Worked example

Read how a full-mark answer is built

A steel rod of diameter 10 mm carries a tensile load of 5 kN.

1 Calculate the stress in the rod. (3 marks)

Full-mark answer each ✓ is one mark

$A = \pi \times 5^2 = 78.5\ mm^2$ ✓. $\sigma = F \div A = 5000 \div 78.5$ ✓ $= 64\ N/mm^2$ ✓.

Why marks are not awarded

"$\sigma = 5 \div 78.5 = 0.064\ N/mm^2$"

2 marks: kN not converted to N.

WE DO — Guided practice

Use the starters and word bank to help you

A 2 m wire stretches by 1.2 mm.

2 Calculate the strain. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which justification scores?
☐ It is strong ☐ It is corrosion resistant, so it won't degrade in sea water ☐ It is good ☐ It is cheap

b) Strain has units of...
☐ N/mm² ☐ mm ☐ None ☐ N

c) A 1152 N force on 36 mm² gives a stress of...
☐ 32 N/mm² ☐ 41 472 N/mm² ☐ 0.031 N/mm² ☐ 1188 N/mm²

4 Key terms: write the letter of the matching description next to each term.

1. **Stress**	____	**A** Name the property and link it to the job.
2. **Strain**	____	**B** $\sigma = F \div A$ (N/mm²)
3. **Justifying a material**	____	**C** $\varepsilon = \Delta l \div l$ (no units)

A hydrofoil mast must be strong, light and resist corrosion in seawater.

5 Choose a suitable material from aluminium, mild steel and carbon fibre, and justify your choice. (2 marks)

During a test, a steel rod stretches 0.6 mm. The strain was 0.0012.

6 Calculate the original length of the rod. (2 marks)

A material has a stress of 64 N/mm² and strain of 3.2 × 10⁻⁴.

7 Calculate Young's modulus and identify the material from the data booklet. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?